Fiscal Year:	FY 2018	**Task Last Updated:**	FY 01/08/2019
PI Name:	Wang, Clay Ph.D.		
Project Title:	Influence of Microgravity on the Production of Aspergillus Secondary Metabolites (IMPAS) - a Novel Drug Discovery Approach with Potential Benefits to Astronauts' Health		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	clayw@usc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	323-442-1670
Organization Name:	University of Southern California		
PI Address 1:	Pharmacology and Pharmaceutical Sciences		
PI Address 2:	1985 Zonal Ave		
PI Web Page:			
City:	Los Angeles	**State:**	CA
Zip Code:	90089-5305	**Congressional District:**	37
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	03/31/2018
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
Contact Email:	kevin.y.sato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 3/31/2018 per F. Hernandez/ARC; original end date was 10/31/2017 (Ed., 8/31/17) NOTE: End date changed to 6/30/2017 per F. Hernandez/ARC; original end date was 10/31/2017 (Ed., 7/28/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Venkateswaran, Kasthuri Ph.D. (NASA Jet Propulsion Laboratory)		
Grant/Contract No.:	NNX15AB49G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The proposed influence of microgravity in the production of Aspergillus secondary metabolites (IMPAS) – a novel drug discovery approach that potentially benefits astronauts' health project will address the effects of microgravity and the space environment on fungal growth and physiological responses in spaceflight. We will expose the well-characterized Aspergillus nidulans fungus under microgravity, which has already been shown to produce biotherapeutic compounds on Earth conditions in support of the development of pharmaceuticals (e.g., anti-osteoporosis agents). When demonstrated for the production of biotherapeutics in space, such an approach will potentially benefit astronauts during longer missions and enhance utilization of the International Space Station (ISS) for future pharmaceuticals development in space. The IMPAS project seeks to answer the follow hypothetical question: Since Aspergillus nidulans exhibited production of antibiotics on Earth, do these fungal cells yield useful biotherapeutic secondary metabolites when grown in stressed microgravity conditions? Metabolomic characterization of both flight and ground control specimens will answer this question, which may lead to the discovery of potentially useful biotherapeutic compounds. The IMPAS team will use the well-characterized model filamentous fungi A. nidulans to test the hypothesis. The team will characterize the profile of wild-type A. nidulans and three different mutant strains. The first is a strain deficient in nonhomologous end joining, an important mechanism for DNA damage repair. The second contains a deletion in a gene involved in the epigenetic regulation of secondary metabolites. The third features an overexpression of the gene for a global regulator of secondary metabolism, LaeA. The team will study the effect of microgravity on the production of fungal metabolites, as well as test the hypothesis that spaceflight alters fungal gene regulation, resulting in changes in metabolite production. Once the fungi are returned to Earth, state-of-the art molecular tools will be used to characterize the "omics" of these organisms. The broad, long-term goal of this project is to study the changes encountered in various aspects of fungal "omics" under microgravity, as well as the production of fungal secondary metabolites. Natural products discovered from traditional discovery programs, such as lovastatin (a polyketide by Aspergillus terreus) and penicillin (a nonribosomally synthesized peptide by Penicillium chrysogenum), have demonstrated that filamentous fungi are a rich source of chemotherapeutic agents against a variety of diseases. Genome sequencing of various natural product-producing organisms have shown that, even in well-characterized organisms like A. nidulans and A. terreus, the biosynthesis pathways of the vast majority of natural products are still unknown, largely because they are silent until specific conditions trigger their expression. Research from Clay Wang's (Principal Investigator) laboratory and others have shown that in filamentous fungi secondary metabolite production is highly sensitive to growth conditions. In addition, the IMPAS team has shown that many secondary metabolism pathways are triggered specifically in harsh or stressful conditions. Thus, hidden within these fungal genomes is a treasure trove of natural products yet to be discovered. The proposed IMPAS project will address the effects of microgravity and the space environment on fungal growth and physiological responses in spaceflight (P1 and P2). The relevant Space Biology guiding questions for the IMPAS project are: CMM-1c (alteration in gravity affects microbial replication including genetic and metabolic regulation); CMM-4 (does the enhanced secondary metabolite production or other genetic changes observed occur in fungal species?); and CMM-5 (physiological responses to spaceflight).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The IMPAS project seeks to answer the follow hypothetical question: Since Aspergillus nidulans exhibited production of antibiotics on Earth, do these fungal cells yield useful biotherapeutic secondary metabolites when grown in stressed microgravity conditions? Metabolomic characterization of both flight and ground control specimens will answer this question, which may lead to the discovery of potentially useful biotherapeutic compounds.
Task Progress:	The influence of International Space Station (ISS) conditions on fungal 'omics' is reported. The investigation included the A. nidulans wild-type and 3 mutant strains, two of which were genetically engineered to enhance secondary metabolite (SM) production. Whole genome sequencing (WGS) revealed that ISS conditions altered the A. nidulans genome in specific regions. In strain CW12001, which features overexpression of the SM global regulator laeA, ISS conditions induced the loss of the laeA stop codon. Differential expression of proteins involved in stress response, carbohydrate metabolic processes, and SM biosynthesis was observed. ISS conditions significantly decreased prenyl xanthone production in the wild-type strain and increased asperthecin production in LO1362 and CW12001, which are deficient in a major DNA repair mechanism. These data provide valuable insights into the adaptation mechanism of A. nidulans to spacecraft environments and present many economic benefits.
Bibliography Type:	Description: (Last Updated: 09/05/2022)
Articles in Peer-reviewed Journals	Guo CJ, Sun WW, Bruno KS, Oakley BR, Keller NP, Wang CCC. "Spatial regulation of a common precursor from two distinct genes generates metabolite diversity." Chem Sci. 2015 Oct 1;6(10):5913-21. Epub 2015 Jul 13. https://doi.org/10.1039/c5sc01058f ; PubMed PMID: 28791090; PubMed Central PMCID: PMC5523082 , Oct-2015
Articles in Peer-reviewed Journals	Romsdahl J, Blachowicz A, Chiang AJ, Chiang YM, Masonjones S, Yaegashi J, Countryman S, Karouia F, Kalkum M, Stajich JE, Venkateswaran K, Wang CCC. "International Space Station conditions alter genomics, proteomics, and metabolomics in Aspergillus nidulans." Appl Microbiol Biotechnol. 2019 Feb;103(3):1363-77. Epub 2018 Dec 12. https://doi.org/10.1007/s00253-018-9525-0 ; PubMed PMID: 30539259 [Note listed previously as Under review for publication to Science Advances in PI's August 2017 report] , Feb-2019
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Articles in Peer-reviewed Journals	Romsdahl J, Blachowicz A, Chiang YM, Venkateswaran K, Wang CCC. "Metabolomic analysis of Aspergillus niger isolated from the International Space Station reveals enhanced production levels of the antioxidant pyranonigrin A." Front Microbiol. 2020 May 21;11:931. https://doi.org/10.3389/fmicb.2020.00931 ; PMID: 32670208; PMCID: PMC7326050 , May-2020

Books/Book Chapters	Blachowicz A, Venkateswaran K, Wang CCC. "Persistence of fungi in atypical, closed environments: Cultivation to omics." in "Microbiology of Atypical Environments. Methods in Microbiology. Vol. 45." Ed. V. Gurtler, J. Trevors. London: Academic Press, 2018. p. 67-86. https://doi.org/10.1016/bs.mim.2018.07.006 , Oct-2018